

DATA 6950 Syllabus Capstone II

Course Information

Course Name and Number: ***DATA-6950 Capstone II***

Classroom: *TBA*

Class Schedule: *TBA*

Lecture/Lab/Total Credits: *3/0/3*

Meet Your Instructor

Instructor Name and Pronouns: *Dr. Weijie Pang (she/her/hers)*

Office Location: *Dobbs Hall 133*

Office Hours: *TBA (will be released by BrightSpace announcement)*

Email Address: *pangw@wit.edu*

Course Description

This course is the second half of the Data Science and Business Analytics capstone sequence. Students complete the project they proposed in Capstone I, setting forth a constructive argument supported by an analysis of existing literature and synthesis of data. Upon completing the project, the students are to communicate their findings effectively to an audience, in oral, visual, and/or in written format.

Course Prerequisites/Corequisites

DATA 6900 Capstone I

Course Learning Outcomes

By the end of this course, students will be able to:

1. Analyze/query data using the Structured Query Language
2. Translate business requirements to an appropriate database model
3. Design a data warehouse using a schema model, such as the star schema

4. Implement a database using a relational database management system

Required Textbooks

Purchase the following required textbooks at the [Wentworth Bookstore](#) located at 103 Ward Street:

- None

Recommended Learning Materials

The following books are recommended, but not required to be purchased:

- Creswell, John W., and J. David Creswell. *Research design: Qualitative, quantitative, and mixed methods approaches*. Sage Publications, 2017. ISBN: 978-1506386706
- Kahneman, Daniel. *Thinking, fast and slow*. Macmillan, 2011. ISBN: 978-0374275631
- Huff, Darrell. *How to lie with statistics*. WW Norton & Company, 1993. ISBN: 978-0-393-09426-8

Instructional Methodologies

This course aims to build a comprehensive capstone project, the students are expected to work independently while keeping in regular contact with the Professor regarding their homework, drafts, reports, and presentations.

Grading Criteria

Item	Weight	Policy
Proposal	5%	Project proposal document needs to be submitted at the beginning of the semester
Project Plan	5%	Project time plan document should include all the necessary tasks and milestones required to successfully complete the project.
Progress Report	30%	All required progress reports must be submitted on time. Progress reports will be evaluated based on completeness, clarity, and demonstrated progress toward project goals.
Participance	10%	Students are expected to attend weekly meetings on time and actively participate in discussions and project activities.

Item	Weight	Policy
Midterm Presentation	10%	Students must present project progress at mid-semester, including project objectives, methodology, completed work, and next steps.
Final Report	25%	An IEEE format final project report and all coding must be submitted at the end of the semester using Latex/Overleaf.
Final Presentation	15%	Students must deliver a final presentation during a public group meeting at the last week in the semester

Attendance Policy

The coursework will take place in an online environment. You will participate in a variety of online activities in which you will earn points toward your final grade. You are expected to contribute constructively to all activities and attend our synchronous sessions. You are expected to attend all of the synchronous and asynchronous sessions. For the asynchronous sessions, attendance may be taken based on your participation in the online activities, such as reviewing the online materials, participating in discussion forums, etc. Unexcused absences from the synchronous meetings will have a negative impact on your attendance and participation grade.

Late Work Policy

There is no makeup for any required submission for any reason. Please bring any such conflicts to the attention of your professors before the beginning of the penultimate week of class (prior to two weeks before the last day of class). Last-minute requests cannot be accommodated.

Successful Students

State how students can be successful in this course. Provided are generic examples. Feel free to add, remove, or customize for your own course:

To be successful in this course:

- *Check the course announcements regularly*
- *Attend the required class meetings*
- *Arrive on time*
- *Participate in in-class activities*
- *Read and respond to email messages as needed*

- Complete assignments by the due dates specified
- Communicate regularly with your instructor and peers
- Create a study schedule to stay on track
- Ask for help early
- Use student support resources such as the [Success Studio](#), [Center for Wellness](#), and [Schumann Library](#)
- Balance mental, physical, and social health to support academic success.

Instructor Promise/Shared Expectations

As your instructor, I believe that every student in my course is capable of greatness. I have high standards for success, and I believe that you can meet them with time, effort, and engagement. I promise to care about your success as a student and your well-being as a person.

Wentworth Grading System

Grade	Weight	Numerical Definition	Definition
A	4.00	93-100	Distinction
A-	3.67	90-92	High Pass
B+	3.33	87-89	Pass
B	3.00	83-86	Pass
B-	2.67	80-82	Provisional
C+	2.33	77-79	Provisional
C	2.00	73-76	Provisional
F	0.00	0-72	No Pass
P	0.00		Pass (for credit)
S	0.00		Satisfactory (no credit)
U	0.00		Unsatisfactory (no credit)
W	0.00		Withdrew
IC	0.00		Incomplete
NR	0.00		Not Reported

Grades of "P" or "F" are awarded to courses with this grading scheme and carry academic credit. "P" or "F" grades do not calculate into the GPA. Wentworth does not offer students the option to audit a course; if a student is granted an exception to this policy the course cannot be converted at any time to a credit-bearing course and will not satisfy a degree requirement.

Add/Drop

Please check the [academic calendar](#) to confirm the add/drop deadline. Add/drop is done online. Courses dropped in this period are removed from your record.

Non-attendance does not constitute dropping a course. If you registered for a course and subsequently withdrew or received a failing grade in its prerequisite, then you must drop that course. In some cases, you will be dropped from that course by the Registrar. However, it is your responsibility to make sure you meet the course prerequisites and to drop a course if you have not successfully completed the prerequisites. Please meet with your student success advisor or major advisor for schedule revision and to discuss the impact of the failed or withdrawn course on your degree status.

Colleges of the Fenway Students

If you are enrolled in this course through COF Cross Registration, please let me know and provide me with your email address so I can be sure that you receive course information in a timely way. Let's also discuss how to access online applications that might be used in the course.

Non-Discrimination Statement

Per Wentworth's Office of Institutional Equity: "Wentworth Institute of Technology values diversity, equity, and inclusion. The University is committed to providing a safe and respectful educational experience and work environment free from discrimination and harassment on the basis of an individual's race, color, religion, gender, age, marital status, national origin, ancestry, alienage, physical or mental disability, sexual orientation, gender identity or expression, genetic information or any other characteristic protected by law."

For more information on reporting, refer to the Office of Institutional Equity section below.

Accessibility Statement

Students with physical, medical, psychiatric, and learning disabilities are eligible to utilize [Student Accessibility Services](#) (a branch of the Success Studio) to arrange for reasonable accommodations and seek ongoing support at any point in their Wentworth career. Please contact AccessibilityServices@wit.edu to begin the accommodation request process.

For more information on accessibility, refer to the Success Studio section below.

Academic Integrity

Members of our community are expected to be honest and forthright in all their academic endeavors. Not only does this add value and credibility to a Wentworth degree, but it also establishes trust and respect as we share knowledge within and beyond our community.

Academic integrity includes:

- Acknowledging the work of others with citations and attributions
- Submitting work that authentically reflects your own knowledge, skills, and abilities – not someone else's
- Collaborating with others on work only when collaborating is an intended part of the activity
- Respecting each other's work, property, and possessions

Learn more about academic integrity by contacting SuccessStudio@wit.edu. [View our Academic Honesty Policy Procedures](#) to learn how Wentworth investigates potential academic integrity violations.

Intellectual Property

Any intellectual property developed by you as part of your work for this course is yours. You own it! [View Wentworth's Intellectual Property policy](#) for details.

Student Support Resources

Wentworth is proud to offer a variety of support resources to enable students to succeed academically, professionally, and personally. Due to the variety of resources available, start at [WhoCanHelp](#) to determine what your needs are and what offices and organizations are best suited to help you. Here is a sample of the student support resources listed on WhoCanHelp:

Success Studio

At Wentworth, we are wholeheartedly committed to student success. In support of this mission, the [Success Studio](#) acts as the central hub dedicated to helping our students thrive through the provision of Advising, Student Accessibility Services, and Tutoring. Questions? Email SuccessStudio@wit.edu.

Center for Wellness

At Wentworth, we value physical, mental, social, and spiritual wellbeing as much as academic success. The [Center for Wellness](#) is comprised of Counseling Services, Health Promotion and Education, as well as Fitness and Wellness Programs. Each provides specific services to our students while working together holistically to improve students' experiences. Need urgent support? [View all crisis hotlines and resources](#).

Douglas D. Schumann Library & Learning Commons

The [Schumann Library](#) provides numerous on-campus and virtual services to the Wentworth community, including an extensive library catalog, research support, study spaces, copyright guides, and more.

Center for Diversity and Global Engagement

The [Center for Diversity and Global Engagement](#) (CDGE) programs at Wentworth promote cross-cultural perspectives, inclusivity, leadership development and personal growth among the Wentworth community, and fosters a welcoming, brave and safe environment for all students with a specific focus on underserved populations and their allies.

Office of Institutional Equity

The [Office of Institutional Equity](#) supports the university in cultivating and maintaining a campus environment that is equitable, inclusive, and accessible. Students, faculty, staff, and guests have a right to be free from discrimination and harassment, including harassment and sexual misconduct. You can report discrimination, harassment, and sexual misconduct to the University by using the [Reporting Bias and Discrimination Form](#) or the [Sexual Misconduct and Sex-Based Discrimination Reporting Form](#).

Course Schedule: Syllabus Outline

A tentative schedule is provided below. The schedule is subject to change as the semester progresses.

Week	Topics	Assignments
1	Project planning	Proposal
2	Project planning	Proposal
3	Project planning	Project plan
4	Research methodology	Progress report I
5	Market research and data collection	
6	Strategy and data analysis, Wellbeing Day – No meeting	Progress report II
7	Strategy and data analysis	
8	Implementing a solution Midterm presentation	Progress report III
9	Spring break – No meeting	
10	Project draft 1	Report draft 1
11	Critique	Progress report IV
12	Implementing a solution	Report draft 2
13	Critique	Presentation slides draft
14	Final Report & Presentation	Final Presentation
15	Final Report & Presentation	Final Report